

Module 4 in-class Coding IV (11/21)

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Java Swing Warehouse Management Lab Assignment

Objective:

The objective of this lab assignment is to create a Java Swing application for managing a warehouse. The program should allow users to check the number of receiving items, enter the price of the item, calculate the total, display the current stock, and release items from the warehouse. Additionally, the program should store and retrieve data from a MariaDB database.

Task Description:

1. Database Setup (15 points):

- Implement the `createTableIfNotExists` method to create a table named `warehouse_data` with the specified columns: `id`, `received_items`, `item_price`, `total_price`, and `remaining_stock`.
- Ensure that the table includes appropriate data types and constraints.

2. GUI Implementation (25 points):

- Implement the Swing GUI to include text fields for entering the number of receiving items and item price, buttons for calculating the total and releasing items, and labels for displaying the total and remaining stock.

3. Database Interaction (30 points):

- Implement the `saveToDatabase` method to insert data into the `warehouse_data` table when items are received.
- Implement the `updateDatabaseForRelease` method to update the remaining stock in the database when items are released.
- Handle database connections and statements appropriately.

4. Calculation and Display (20 points):

- Implement the `calculateTotal` method to calculate the total price based on the entered receiving items and item price.

- Implement the `releaseItems` method to update the remaining stock when items are released.
- Display the calculated total and remaining stock on the GUI.

5. Code Structure and Readability (10 points):

- Ensure the code follows good practices, including proper indentation and meaningful variable/method names.
- Include comments where necessary to explain the functionality of methods and sections of code.

Grading Rubric:

• Database Setup (15 points):

- Successful creation of the `warehouse_data` table with the correct structure: 5 points.
- Appropriate data types and constraints: 5 points.
- Proper error handling for database operations: 5 points.

• GUI Implementation (25 points):

- Presence of GUI components (text fields, buttons, labels): 5 points.
- Proper layout (e.g., GridLayout) and organization of GUI elements: 5 points.
- Event handling for button clicks: 5 points.
- User-friendly design and clear component labels: 5 points.
- Proper error handling for user inputs: 5 points.

• Database Interaction (30 points):

- Successful insertion of data into the `warehouse_data` table: 10 points.
- Successful updating of the remaining stock in the database: 10 points.
- Proper error handling for database operations: 5 points.
- Efficient handling of database connections and statements: 5 points.

• Calculation and Display (20 points):

- Successful calculation of the total price: 10 points.
- Successful updating of remaining stock: 10 points.

- **Code Structure and Readability (10 points):**

- Consistent indentation and formatting: 5 points.
- Meaningful variable/method names: 5 points.

**** Skeleton Code ****

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.sql.*;

public class WarehouseManager {

    private static final String DB_URL = "jdbc:mariadb://localhost:3306/warehouse_db";
    private static final String DB_USER = "your_username";
    private static final String DB_PASSWORD = "your_password";

    private int currentStock = 0;
    private double totalPrice = 0;

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            WarehouseManager warehouseManager = new WarehouseManager();
            warehouseManager.createAndShowGUI();
        });
    }

    private void createAndShowGUI() {
        JFrame frame = new JFrame("Warehouse Manager");
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        JPanel panel = new JPanel();
        frame.add(panel);
```

```
        placeComponents(panel);

        frame.setSize(400, 200);
        frame.setVisible(true);
    }

    private void placeComponents(JPanel panel) {
        // TODO: Implement GUI components (text fields, buttons, labels)

        JButton calculateButton = new JButton("Calculate Total");
        JLabel totalLabel = new JLabel("Total: $0.0");

        JButton releaseButton = new JButton("Release Items");
        JLabel remainingStockLabel = new JLabel("Remaining Stock: 0");

        // TODO: Add components to the panel and set layout

        calculateButton.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                // TODO: Implement calculation and display logic
            }
        });

        releaseButton.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                // TODO: Implement release items logic
            }
        });
    }

    private void calculateTotal(String receivingItems, String price, JLabel totalLabel) {
        // TODO: Implement calculation and display logic
    }
}
```

```
}  
  
private void releaseItems(String releasingItems, JLabel remainingStockLabel) {  
    // TODO: Implement release items logic  
}  
  
private void saveToDatabase(int received, double itemPrice) {  
    // TODO: Implement database insertion logic  
}  
  
private void updateDatabaseForRelease(int released) {  
    // TODO: Implement database update logic  
}  
}
```

Points 100

Submitting a file upload

Due	For	Available from	Until
Nov 24, 2025	Everyone	-	Nov 24, 2025 at 11:59pm